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# Gene Therapy

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# Gene Therapy

- Gene therapy is a technique for correcting defective genes responsible for disease development.
- Researchers may use one of several approaches for correcting faulty genes:
  1. A normal gene may be inserted into a nonspecific location within the genome to replace a nonfunctional gene - most common.
  2. An abnormal gene could be swapped for a normal gene
  3. The abnormal gene could be repaired through selective reverse mutation, which returns the gene to its normal function.
  4. The regulation (the degree to which a gene is turned on or off) of a particular gene could be altered.

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# Mechanism

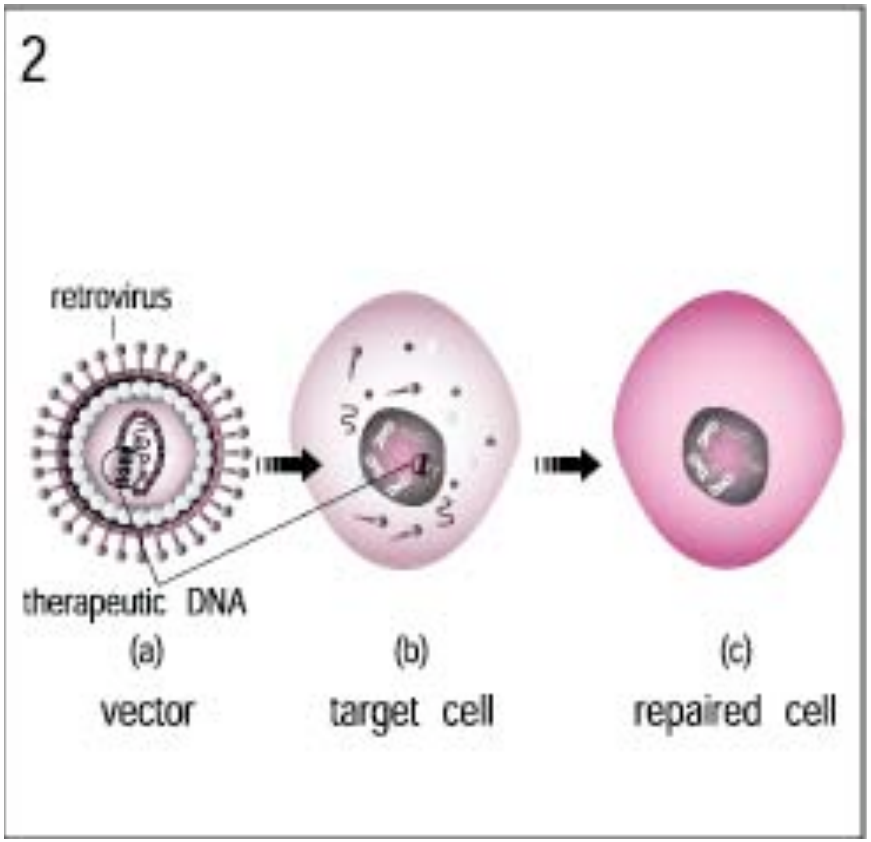
- A carrier molecule called a vector must be used to deliver the therapeutic gene to the patient's target cells.
- Currently, the most common vector is a virus that has been genetically altered to carry normal human DNA.
- Viruses have evolved a way of encapsulating and delivering their genes to human cells in a pathogenic manner.
- Scientists have tried to take advantage of this capability and manipulate the virus genome to remove disease-causing genes and insert therapeutic genes.

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# Mechanism

- Target cells such as the patient's liver or lung cells are infected with the viral vector.
- The vector then unloads its genetic material containing the therapeutic human gene into the target cell.
- The generation of a functional protein product from the therapeutic gene restores the target cell to a normal state.




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# Virual Vectors

- **Retroviruses** - A class of viruses that can create double-stranded DNA copies of their RNA genomes. These copies of its genome can be integrated into the chromosomes of host cells. Human immunodeficiency virus (HIV) is a retrovirus.
- **Adenoviruses** - A class of viruses with double-stranded DNA genomes that cause respiratory, intestinal, and eye infections in humans. The virus that causes the common cold is an adenovirus.
- **Adeno-associated viruses** - A class of small, single-stranded DNA viruses that can insert their genetic material at a specific site on chromosome 19.
- **Herpes simplex viruses** - A class of double-stranded DNA viruses that infect a particular cell type, neurons. Herpes simplex virus type 1 is a common human pathogen that causes cold sores.

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# Other Mechanisms of Transfer

- Besides virus-mediated gene-delivery systems, there are several nonviral options for gene delivery.
- The simplest method is the direct introduction of therapeutic DNA into target cells. This approach is limited in its application because it can be used only with certain tissues and requires large amounts of DNA.
- Another nonviral approach involves the creation of an artificial lipid sphere with an aqueous core. This liposome, which carries the therapeutic DNA, is capable of passing the DNA through the target cell's membrane.

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# A therapeutic 47th Autosome?

- Researchers also are experimenting with introducing a 47th (artificial human) chromosome into target cells.
- This chromosome would exist autonomously alongside the standard 46 --not affecting their workings or causing any mutations.
- It would be a large vector capable of carrying substantial amounts of genetic code, and scientists anticipate that, because of its construction and autonomy, the body's immune systems would not attack it.
- A problem with this potential method is the difficulty in delivering such a large molecule to the nucleus of a target cell.

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# Affecting Factors

**Short-lived nature of gene therapy**

- Before gene therapy can become a permanent cure for any condition, the therapeutic DNA introduced into target cells must remain functional and the cells containing the therapeutic DNA must be long-lived and stable.
- Problems with integrating therapeutic DNA into the genome and the rapidly dividing nature of many cells prevent gene therapy from achieving any long-term benefits.
- Patients will have to undergo multiple rounds of gene therapy.

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# Affecting Factors

**Immune response**

- Anytime a foreign object is introduced into human tissues, the immune system is designed to attack the invader.
- The risk of stimulating the immune system in a way that reduces gene therapy effectiveness is always a potential risk.
- Furthermore, the immune system's enhanced response to invaders it has seen before makes it difficult for gene therapy to be repeated in patients.

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# Affecting Factors

**Problems with viral vectors**

- Viruses, while the carrier of choice in most gene therapy studies, present a variety of potential problems to the patient --toxicity, immune and inflammatory responses, and gene control and targeting issues.
- In addition, there is always the fear that the viral vector, once inside the patient, may recover its ability to cause disease.

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# Affecting Factors

**Multigene disorders**

- Conditions or disorders that arise from mutations in a single gene are the best candidates for gene therapy.
- Unfortunately, some the most commonly occurring disorders, such as heart disease, high blood pressure, Alzheimer's disease, arthritis, and diabetes, are caused by the combined effects of variations in many genes.
- Multigene or multifactorial disorders such as these would be especially difficult to treat effectively using gene therapy.

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# Ethics & Gene Therapy

1. What is normal and what is a disability or disorder, and who decides?
2. Are disabilities diseases? Do they need to be cured or prevented?
3. Does searching for a cure demean the lives of individuals presently affected by disabilities?
4. Is somatic gene therapy (which is done in the adult cells of persons known to have the disease) more or less ethical than germline gene therapy (which is done in egg and sperm cells and prevents the trait from being passed on to further generations)? In cases of somatic gene therapy, the procedure may have to be repeated in future generations.
5. Preliminary attempts at gene therapy are exorbitantly expensive. Who will have access to these therapies? Who will pay for their use?

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